

- 1) Direct form, cascade form, parallel form
- 2) Analysis of difference equation of LTI system

$$y(n) + 3y(n+1) + y(n) - x(n) = 0$$

$$x(n) = u(n+1) - u(n)$$

initial conditions are given

Find $y(n)$?

$$1) \frac{Y(z)}{X(z)}$$

$$Y(z) =$$

$$y(n)$$

3) Jury's test or Bilinear transformation
(Row with an

II
1) D.F.S.

2) Frequency response of simple discrete time

$$H(z) = ?$$

$$H(e^{j\omega}) = \text{Frequency response (1)}$$

Module-4

- 1) Sampling theorem - proof, aliasing.
- 2) ZOH or FOH.
- 3) Discrete convolution - properties.

$$\mathcal{L}^{-1} \{Y(s)\} = ?$$

$H(j\omega)$ = Frequency response.

Module-II

1) Transfer function of LTI system - Electrical Translational and Rotational.

Force voltage analogy

Force current analogy.

2) Fourier series (Trigonometric or exponential)

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magnitude spectrum

frequency spectrum

To find frequency response - find

3) Energy spectral density - Fourier Transform
Power spectral density

Module I

1) Types of properties of non linear system.

2) Basic operations on signals.

3) check. Linearity, Causality, Dynamic, Time variation

4) Classification of signals - Power, Energy

5) Convolution Integral with impulse response

6) Random signals and processes